



$$P = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

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$V_{\alpha\beta} - V_{\alpha\beta}$ (静止坐标系的数学建模)

$$\begin{cases} V_d = R i_d + L_d \frac{d i_d}{dt} - \omega L_q i_q \\ V_q = R i_q + L_d \frac{d i_q}{dt} + \omega L_d i_d + p f \end{cases}$$

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix}, \quad i_{\alpha\beta} = P^T i_{dq}$$

$$V_{\alpha\beta} = P^T V_{dq}$$

$$\frac{d}{dt}(P^T i_{dq}) = P^T i_{dq} + P^T \frac{d i_{dq}}{dt}$$

1. 电压的矢矩阵形式

$$\therefore V_{dq} = R \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \frac{d i_{dq}}{dt} + \omega \begin{bmatrix} 0 & -L_q \\ L_d & 0 \end{bmatrix} i_{dq} + \omega \begin{bmatrix} 0 \\ p f \end{bmatrix}$$

2. 等式左侧乘以 P^T

$$P^T V_{\alpha\beta} = R P^T i_{dq} + P^T \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \frac{d i_{dq}}{dt} + \omega P^T \begin{bmatrix} 0 & -L_q \\ L_d & 0 \end{bmatrix} i_{dq} + \omega P^T \begin{bmatrix} 0 \\ p f \end{bmatrix}$$

$$\frac{d i_{dq}}{dt} = P i_{\alpha\beta} + P \frac{d i_{\alpha\beta}}{dt}$$

3. 定义 $\alpha\beta$ 坐标系时变电感矩阵

$$\therefore P_{\alpha\beta} = P^T P \frac{d i_{dq}}{dt}$$

$$P_{dq} = L_{dq} i_{dq}$$

$$i_{dq} = P^T i_{\alpha\beta} \cdot P i_{\alpha\beta}$$

$$P_{\alpha\beta} = P^T L_{dq} i_{dq} = P^T L_{dq} P i_{\alpha\beta}$$

$$\therefore L_{\alpha\beta} = P^T L_{dq} P$$



$$L_a = \cos\theta \cdot L_d \cos\theta - \sin\theta (-L_q \sin\theta)$$

$$= L_d \cos^2\theta + L_q \sin^2\theta \quad \cos^2 x = \frac{1+\cos 2x}{2} \quad \text{Memo No.}$$

$$L_a = \frac{L_d+L_q}{2} + \frac{L_d-L_q}{2} \cos 2\theta \quad \sin^2 x = \frac{1-\cos 2x}{2} \quad \text{Date } / /$$

$$L_{a\beta}(\theta_e) = P^T \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} P = \begin{bmatrix} L_a & L_{a\beta} \\ L_{a\beta} & L_\beta \end{bmatrix} \quad L_{a\beta} = \cos\theta \cdot L_d \sin\theta - \sin\theta \cdot L_q \cos\theta$$

$$= (L_d - L_q) \sin\theta \cos\theta$$

$$L_a = \frac{L_d+L_q}{2} + \frac{L_d-L_q}{2} \cos 2\theta$$

$$\sin 2\theta = 2 \sin\theta \cos\theta$$

$$L_{a\beta} = -\frac{L_d-L_q}{2} \sin 2\theta$$

$$L_{a\beta} = -\frac{L_d-L_q}{2} \sin 2\theta$$

IPMSM $L_d \neq L_q$, $L_a, L_{a\beta}$ 随转子角度变化, 因而存在 $\frac{dL_a}{dt}$, $\frac{dL_{a\beta}}{dt}$

4. 永磁同步电机模型

$$W_e P^T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = W_e P^T \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix}$$

$$\therefore \begin{bmatrix} v_a \\ v_\beta \end{bmatrix} = \begin{bmatrix} R + \frac{d}{dt} L_a & \frac{d}{dt} L_{a\beta} \\ \frac{d}{dt} L_{a\beta} & R + \frac{d}{dt} L_\beta \end{bmatrix} \begin{bmatrix} i_a \\ i_\beta \end{bmatrix} + W_e P^T \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix}$$

$$1. \text{IPMSM: } L_d \neq L_q$$

$L_a, L_{a\beta}$ 是 $\theta_e(t)$ 的函数, 随转子位置周期性变化 (2 电角步频率), 是时变矩阵 $\frac{dL_a}{dt}, \frac{dL_{a\beta}}{dt}$

$$2. \text{SPMSM (表贴永磁电机): } L_d = L_q = L_s$$

$$L_a = L_s, L_{a\beta} = 0$$

$$L_{a\beta} = \begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix}$$

αβ 坐标系: IPMSM 的电机参数矩阵随转子位置周期性变化, 是时变矩阵, 模型更复杂。这也是工程上很少直接在 αβ 坐标系建立 IPMSM 模型的根本原因

